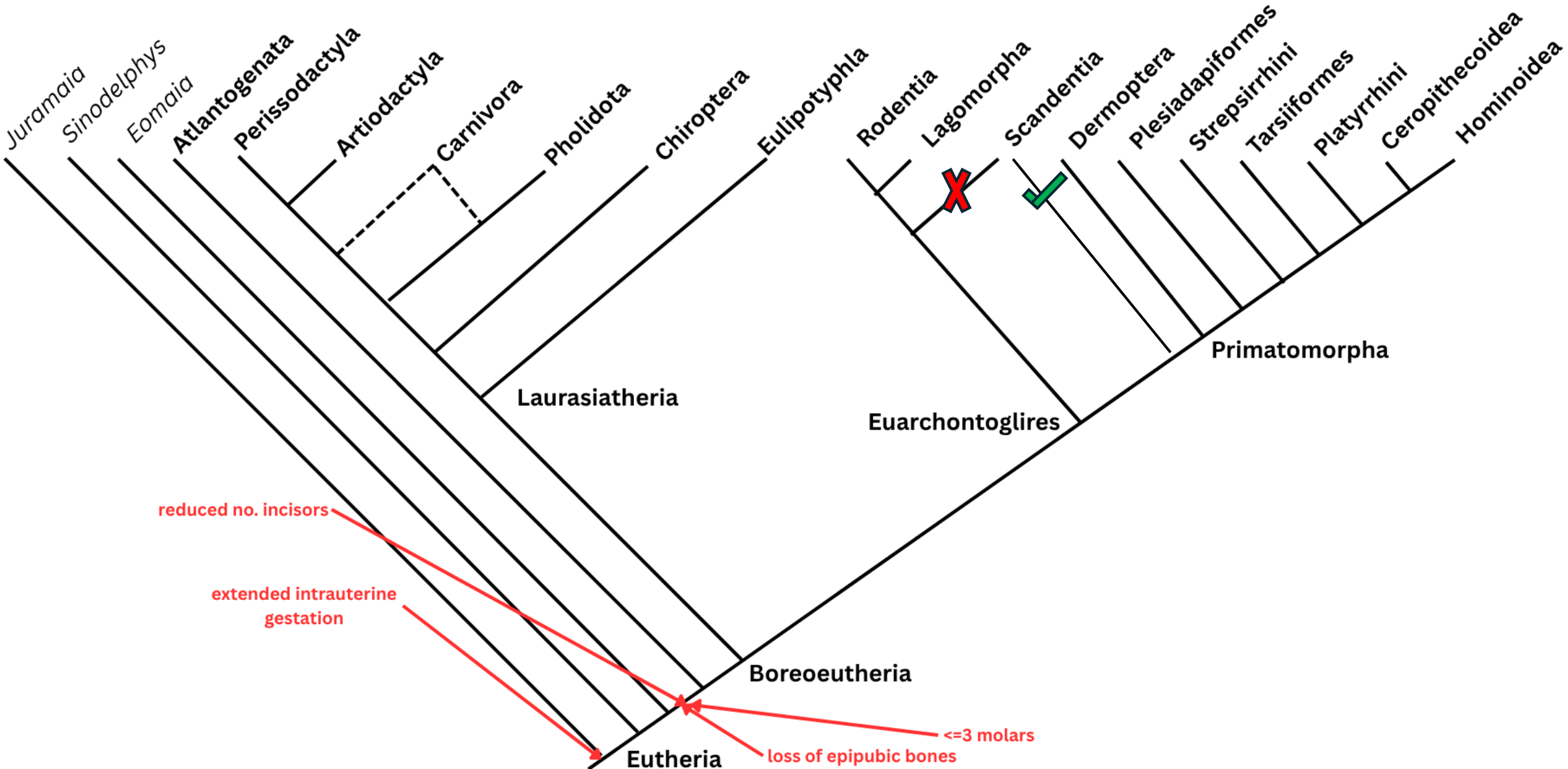
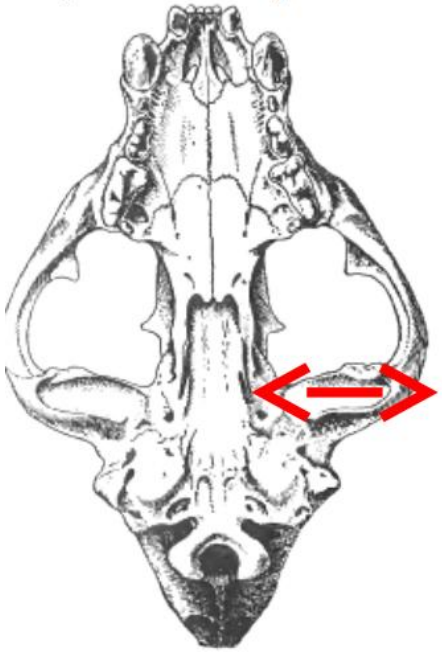
Boreoeutheria



Defining anatomy of Glires

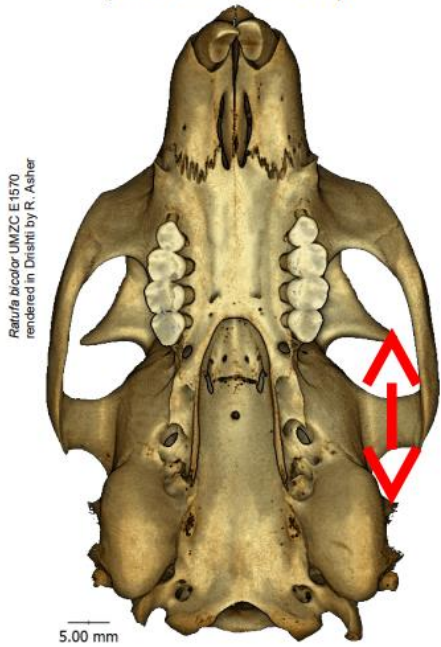
- Enlarged ant. incisor (i2) is deciduous
- Main incisor is curved, enlarged, and ever-growing
- Enamel is restricted to anterior band allowing for self-sharpening
- Diastema from lacking canines and anterior premolars
- Antero-posterior (palinal/"propalinal") jaw movement (bilateral in other mammals)
- Masseter larger than temporalis - more force applied through incisors.

bilateral
(side-to-side)

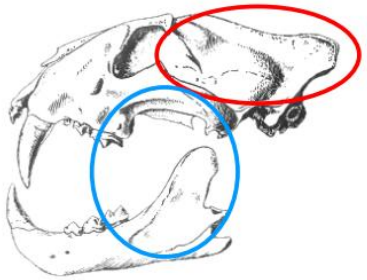


Panthera leo

propalinal
(front-to-back)

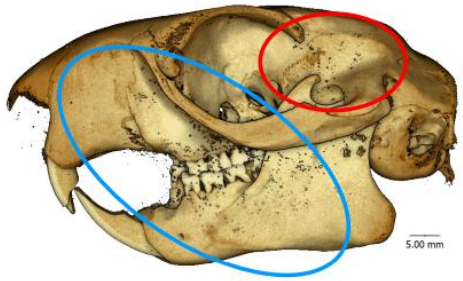


Ratufa bicolor



V. G. Heptner, et al. (1988) Mammals of the Soviet Union, vol. 2, part 2
commons.wikimedia.org/wiki/File:MSU_V2P2_-_Panthera_tigris_altaica_skull.png

masseter and temporalis, both large
- muscle scar on cheekbone
- large lever on jawbone (coronoid process)



Ratufa bicolor UMZC E1570
rendered in Drishti by R. Asher

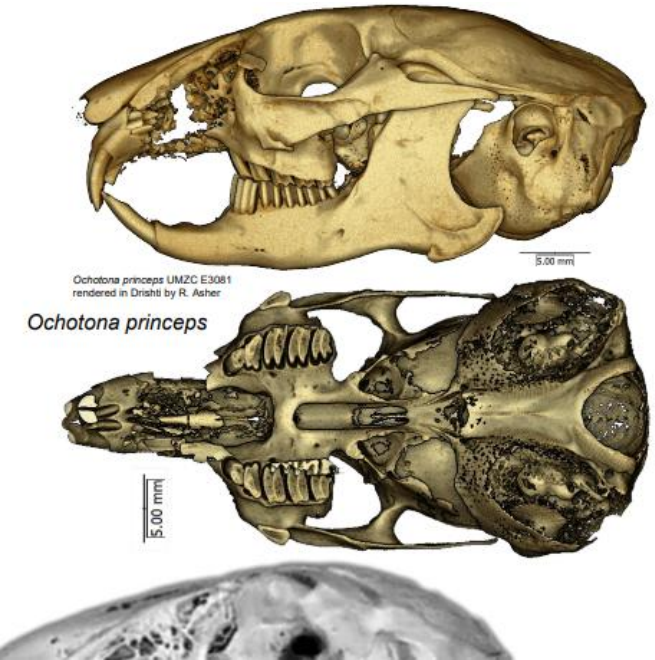
masseter larger than temporalis
- big muscle scar on cheekbone
- small lever on jawbone (coronoid process)

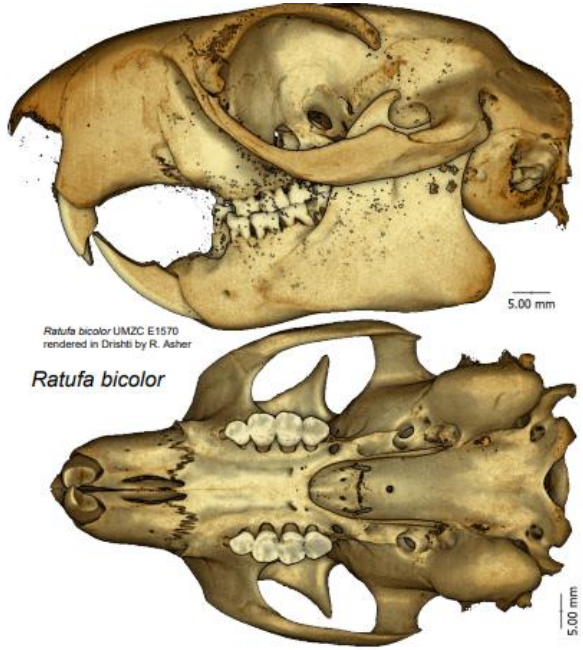
Rodents

- Un-fenestrated rostral maxilla
- No I3
- Larger coronoid process
- Rooted cheek teeth
- Solid Basicranium
- Longer tails (more caudal vertebrae)

Lagomorphs

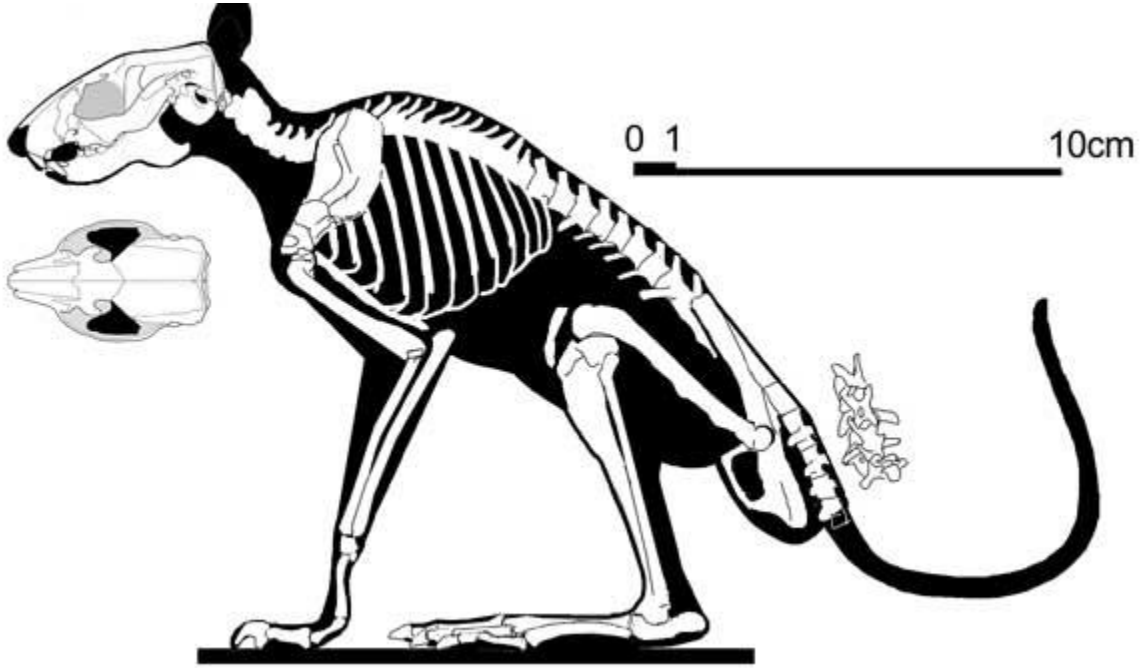
- Fenestrated rostral maxilla
- Additional incisor (i3)
- Reduced coronoid process
- Unrooted cheek teeth
- Basicranial flexion
- Reduced caudal vertebrae number





Lagomorpha

- Split into Ochotonidae (Pikas) & Leporidae (Rabbits and Hares)
- Gomphos elkema – Eocene stem lagomorph
- Long metatarsus – lagomorph trait
- Lagomorph-like calcaneal morphology
- 2 upper incisors – lagomorph trait
- Tail/more caudal vertebrae – rodent trait
- Rooted molars – rodent trait
- Large coronoid process – rodent trait

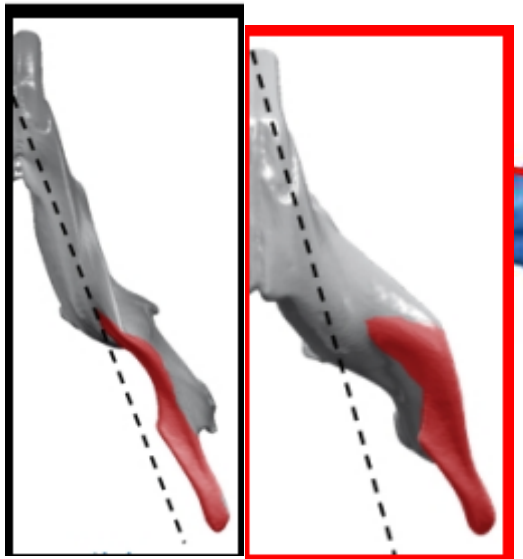


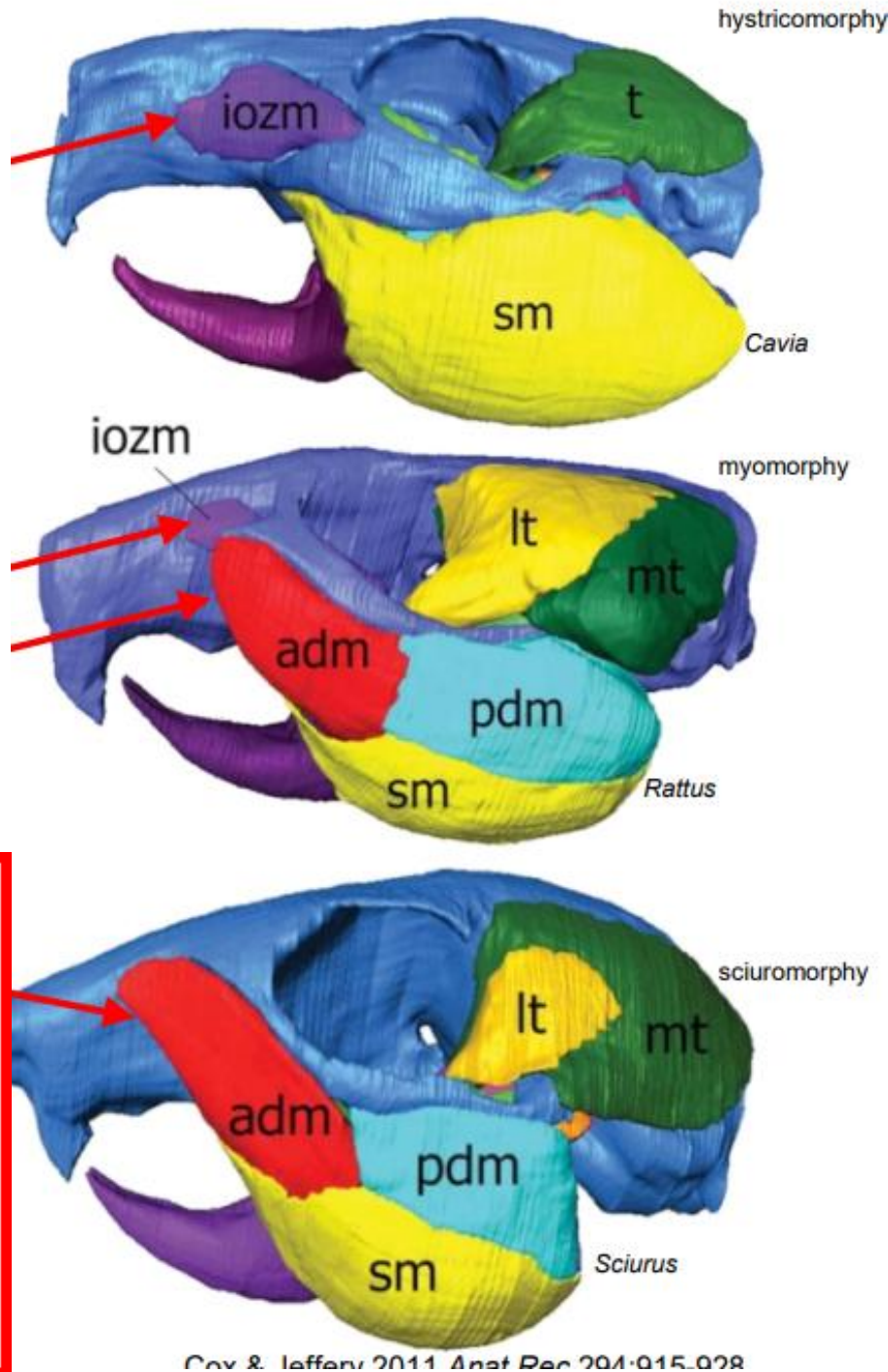
Rodentia (~2000 sp)

Sciuromorpha (squirrels)
- Small infraorbital foramen
- Large muscle scar (zygomatic plate) on or above infraorbital foramen
- Anatomical **sciuromorphy** – masseter slides up anterior zygoma, no muscle through infraorbital foramen
- **Sciurognathy** – angular process in line with incisor alveolus + digastrics contact anteriorly

Ctenohystrica (Guinea pigs etc)
- Large infraorbital foramen
- Muscle scar (zygomatic plate) under & behind infraorbital foramen – essentially NO zygomatic plate
- Anatomical **hystricomorphy** – masseter traverses infraorbital foramen, no zygomatic plate
- **Histricognathy** – angular process far lateral to incisor alveolus + digastrics are separate

Myomorpha (Mice, beavers, etc)
- Large keyhole-shaped infraorbital foramen
- Small muscle scar (zygomatic plate) on or above infraorbital foramen
- Anatomical **myomorphy** – masseter traverses infraorbital foramen AND slides up anterior zygoma (zygomatic plate).
- **Sciurognathy** – angular process in line with incisor alveolus + digastrics contact anteriorly





Cox & Jeffery 2011 Anat Rec 294:915-928

Laonastes

- Sciurognathy – Angular process in-line with incisor alveolus
- Sciurognathy – Digastrics contact anteriorly
- Hystricomorphy – Masseter through infra-orbital foramen

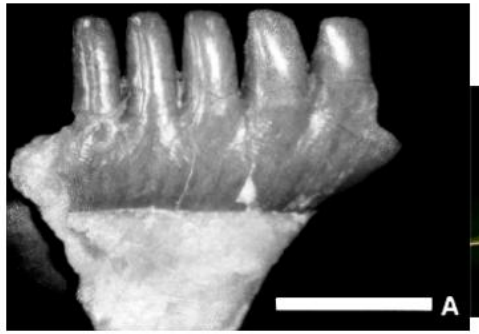
- Partial mitDNA sequencing initially interpreted Laonastes as a hystricognath/ctenohystrica (Guinea-pig-like rodent clade)
- Morphology suggests position outside this clade/Laonastes is like a mammalian coelacanth
- Eventual conclusion was Laonastes was at the very base of Ctenohystrica – closer to gundis than guinea pigs.
- They form clade with Gundis (Ctenodactylidae) sister to Hystricognathi.
- It is anatomically hystricomorph with a sciurognath lower jaw.

Dermoptera (colugos/flying lemurs)

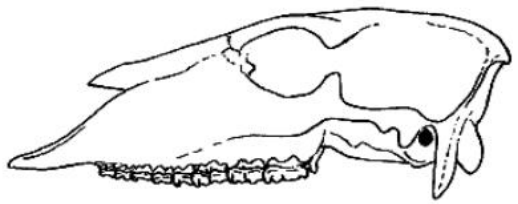
- Typically positioned as the earliest diverging member of primatomorpha.
- Two genera each with 1 species

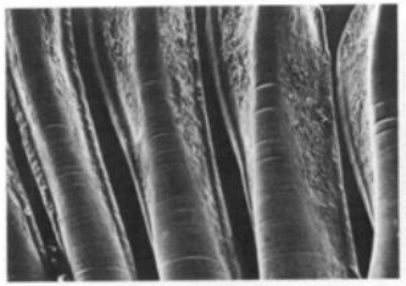
Anatomy

- Patagium (gliding membrane)
- Comb-like lower incisors
- Reduced anterior upper teeth
- Multi-tined grooming incisor
- Common in Borneo



Antilohyrax (Rasmussen & Simons 2000)



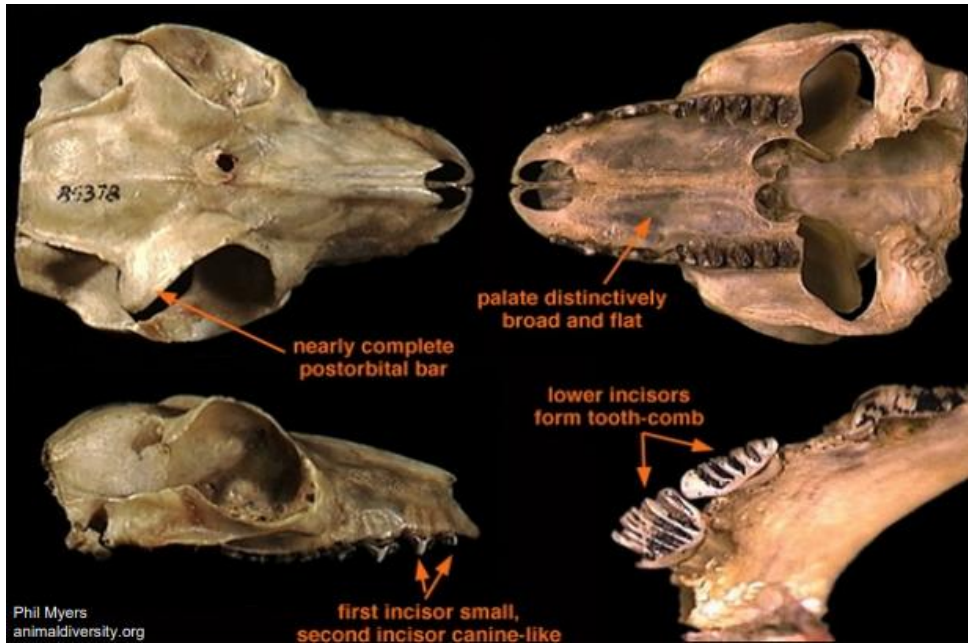


Aimi & Inagaki 1988 J. Mammalogy 69:138

1 mm

1mm

Cynocephalus



Scandentia (tree shrews)

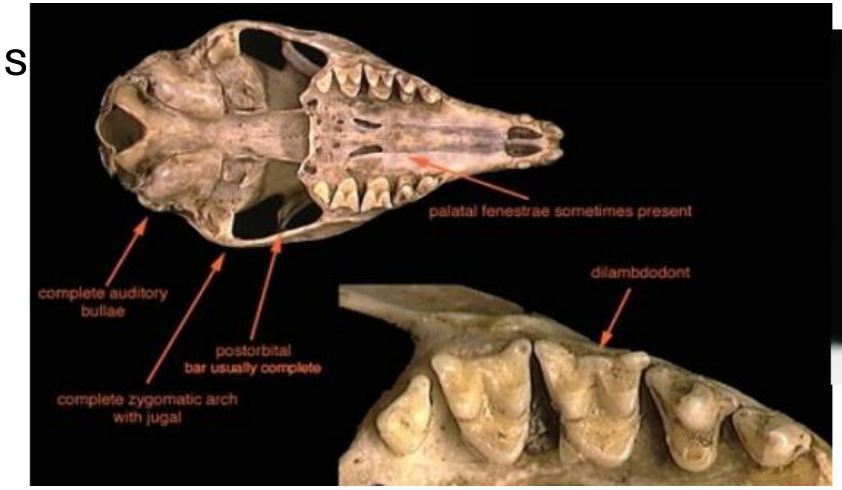
- We don't really know what these are – in older literature were considered primates/primatomorphs because of postorbital bar
- Dermopterans used to be considered chiropteran relatives and scandentians primates.
- This was questioned and dermopterans were found as sister to primates but scandentians moved from primates themselves, to rodents, to sister to glires, to sister to dermopterans.
- The seemingly correct arrangement these days is Scandentia as sister to primatomorpha (incl. Dermoptera).

Anatomy:

- Postorbital bar
- Elongate comb-like incisors
- W-shaped dilambodont molars
- No hypocone
- Bony tubules enclosing middle ear arteries
- Ento/ectotympanic bulla

QUESTION:

- Borneo has a lot of flying/gliding mammals native to it – why?
- Why did primates never take to the skies?



Primates

all zeugopodal elements remain unfused - locomotor flexibility

2,1,3,3 dentition
lateral ext nares
prehensile tail
zygomatic-parietal contact

higher molar crown relief
longer mediodistal distance
wider central basin
thick enamel
terrestrial phase

"Prosimian"

"Anthropoid/Simian"

Glires
Scandentia
Tree Shrews
Dermoptera
Colugos
Plesiadapiformes
Adapidae
Lemuroids
Lorisoids
Omomyidae
Tarsiiformes
Platyrrhini
Oligopithecidae
Propilopithecoidea
Pliopithecoidea
Saadanius
Cerepithecoidea
Kamoyapithecus
Proconsulidae
Hylobatidae
Gigantopithecus
Ponginae
Pierolapithecus
Gorillini
Ouranopithecus
Pan
Graecopithecus
Hominina

asian radiation

Strepsirrhini

Hominini

Homininae

african radiation

Hominidae

Homininoidea

trichromatic vision

european radiation

Catarrhini

2,1,2,3 dentition
ventral nares
frontal-sphenoid contact

grooming claw
tapetum lucidum - eye shine
tooth comb
bicornate uterus
insectivory
monomorphic (sexually)

Simiiformes

Haplorhini

halporhini - dry nose
sexually dimorphic
frugivory/folivory
post-orbital wall

post-orbital bar

molars with
low cusps

grasping hallux
and pollex

nails

conservative
dental formula

Primates

anterior-facing orbits

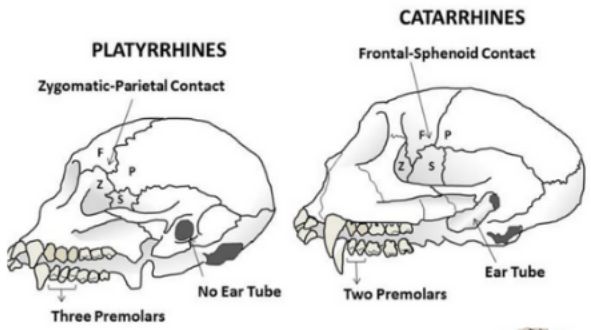
short snout

ancestrally
pentadactyl

Primatomorpha

Euarchonta

Euarchontoglires



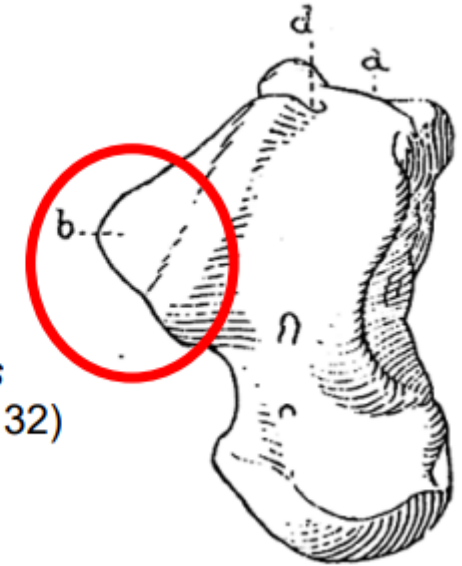
Primates (strepsirhini)

Anatomy

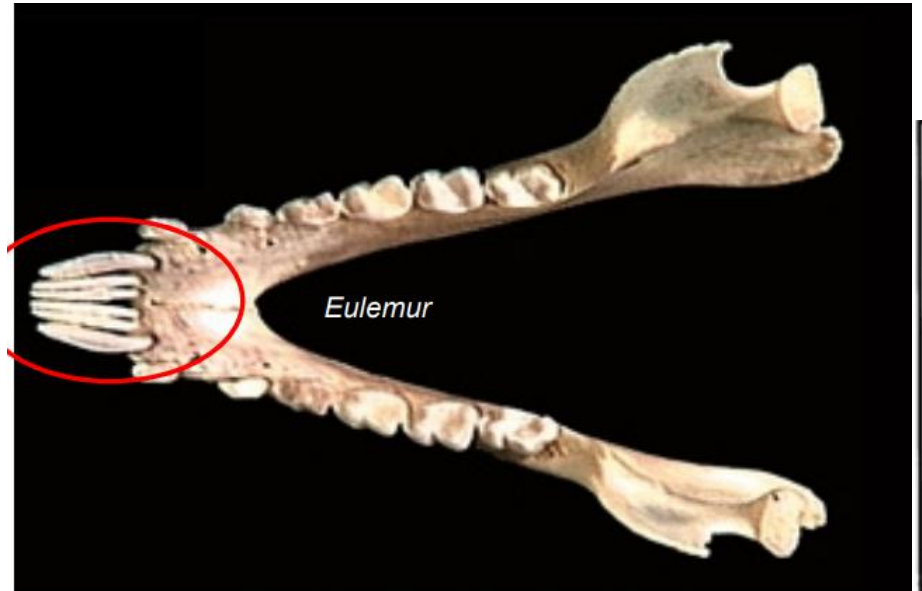
- Tooth comb
- Grooming claw on pedal (foot) digit II – same as dromaeosaurid raised toe claw
- Sloping fibular facet on astragalus
- Postorbital bar – like thylacosmilus
- Wet nose/rhinarium continuous with mouth

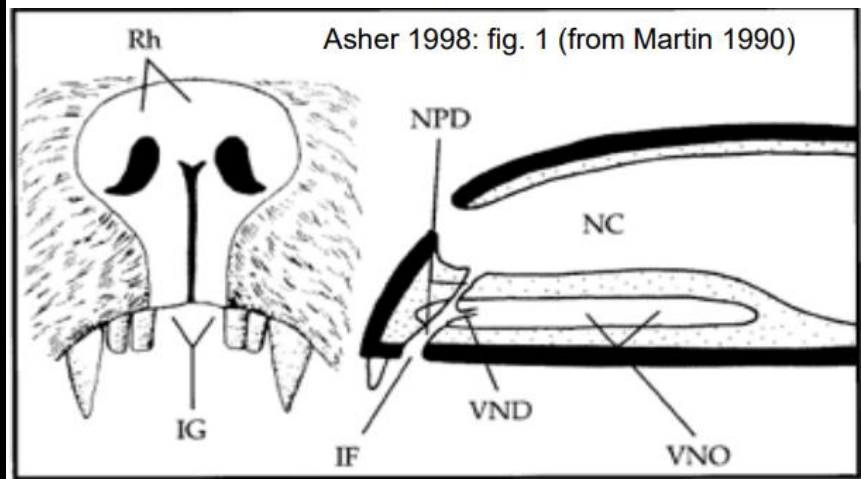
VNO (vomeronasal organ)

- Pheromone sensing organ above palate, usually opens into palate in association with incisive foramen
- Rhinarium connects with vomeronasal organ to better collect olfactory cues/signals
- Flehmen response



Adapis parisiensis
(Stehlin 1916: fig. 32)





Primates (haplorhini)

Shared Anatomy (Tarsiids + Anthropoids)

- Post-orbital closure
- Lack of tapetum lucidum (reflective cells in back of retina used for night-vision)

Anatomy shared by tarsiers with strepsirhines

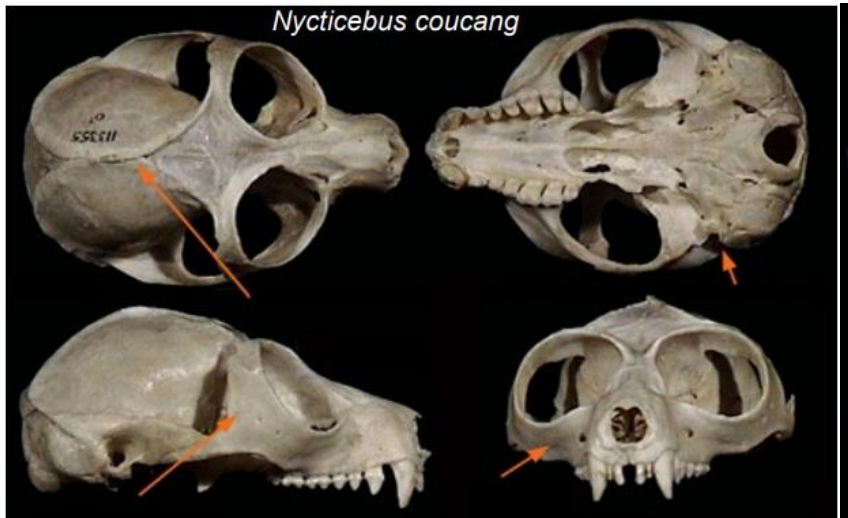
- Unfused mandibular symphysis
- Unfused frontal bone
- Grooming claws

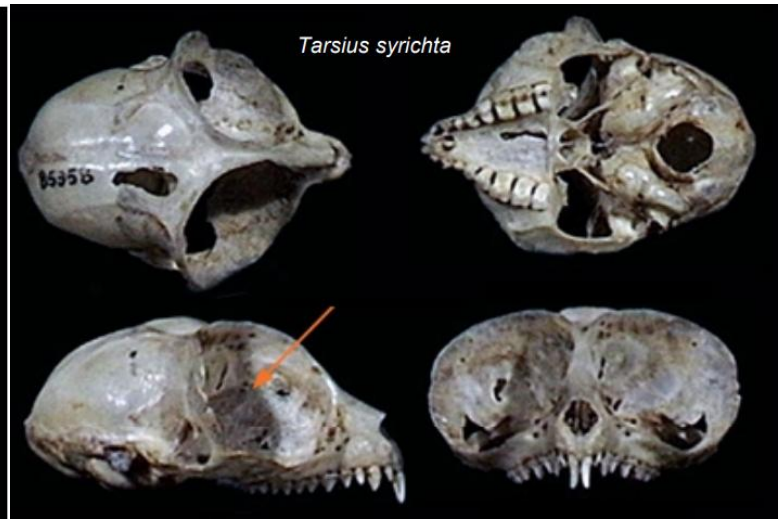
Anatomy unique to Anthropoids

- Fused mandibular symphysis
- Fused frontal bone
- Lack of grooming claws

VNO (vomeronasal organ)

- Despite their dry nose, the VNO still occasionally has a role to play in pheromone sensing
- It's typically lost/redundant in trichromatic primates as visual cues become the primary medium for communication.
- It's even more lost in apes as auditory signals/cues convey yet more information, rendering the use of pheromones relatively obsolete.
- In humans, with language, the vomeronasal organ is completely obsolete. It still forms but all of the 10 (wow) genes responsible are pretty much non-functional.





Primates (Anthropoidea)

- Split into two major clades, catarrhine & platyrrhinni

Platyrrhines:

- Still have vomeronasal organ and dichromatic vision (with exception of howler monkeys (Alouetta).
- HOWEVER, it's on the x chromosome so males are colourblind and females are "often" trichromats)

Catarrhine:

- Split into "old world" monkeys (ceropithecoidea) & apes (homininoidea)
- Old world monkeys likely had a terrestrial/non-arboreal phase in their evolution, so their hands and feet are actually quite well adapted to terrestrial habits.
- Any of you who have seen old world monkeys and apes together will have noticed how much better the apes are at being arboreal.
- Many old-world monkeys are still primarily terrestrial – such as baboons and mandrils.
- A key adaptation of apes is the lack of a tail and highly dexterous hands able to negotiate the canopy with ease.

Howler monkey

Spider monkey

Platyrrhines

Catarrhines

Orangutan (Ape)

Baboon ("Old world")

Apes (Hominoidea)

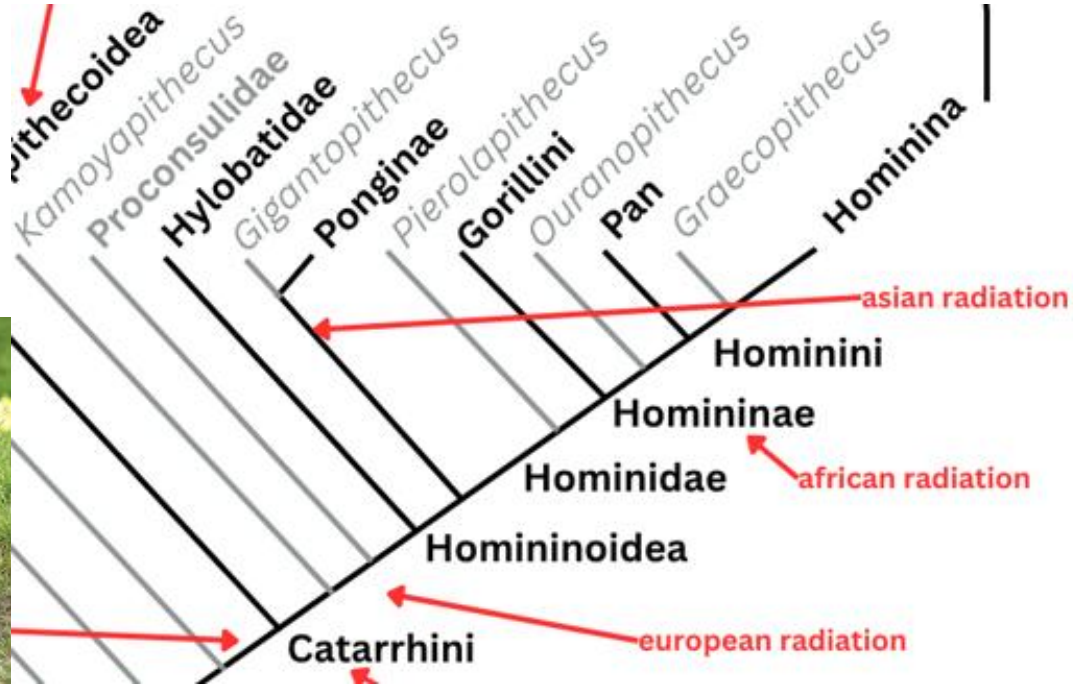
Gibbons
(Hylobactidae)

- Apes appear to have originated in Europe when it was covered in forests during the early Miocene.
- Gibbons (lesser apes) just left for Asia seemingly pretty early – we don't have a good fossil record of them.
- As the earth cooled and the forests retreated south, the great ape lineages (Hominidae) followed eventually splitting into two major clades, the African (Homininae) and Asian clades (Ponginae).
- While Ponginae has little fossil record other than Gigantopithecus, Homininae has a rich fossil record – almost exclusively within Hominina – our ancestors.

Orangutans
(Hominidae, Ponginae)

Humans
(Hominina, Homo)

Chimpanzees & Bonobos
(Hominini, Pan)



Gorillas
(Homininae, Gorillini)

Hominoidea

Locomotion

Gibbons – brachiation (arm only tree arboreality)

Orangutans – quadrumanal climbing (4 limbed climbing)

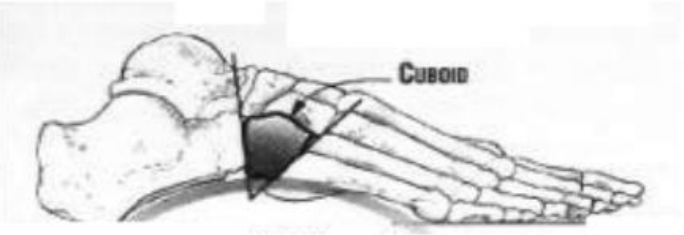
Chimps + Gorillas – knuckle-walking (likely independently evolved)

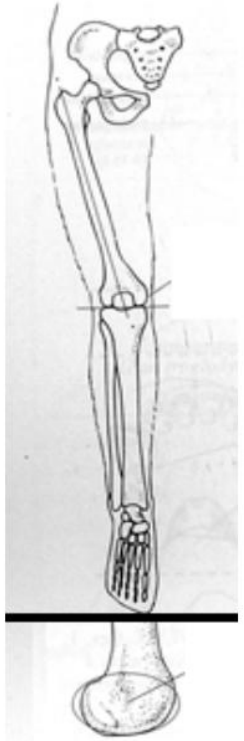
Humans – habitual bipedalism

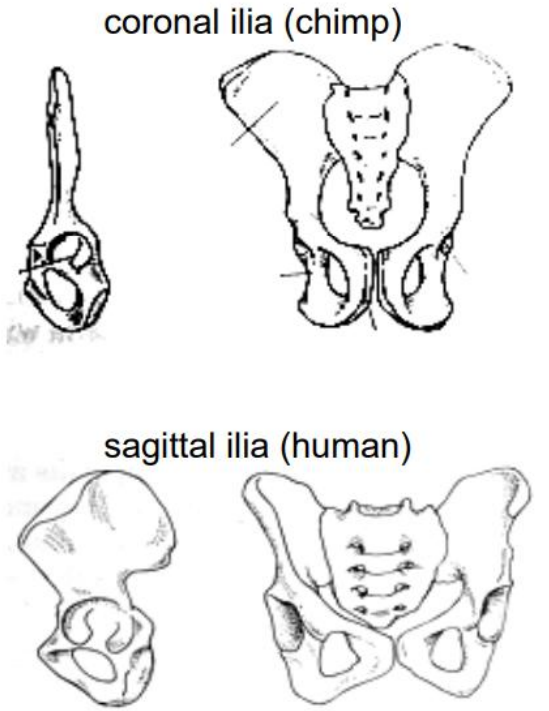
Pretty much all groups are facultative bipeds though.

Osteology of habitual/obligate bipedalism:

- Keystone cuboid
- Straight pedal digit I
- Valgus Knee
- Oval femoral condyles
- Sagittal ilium







Hominini

“Ardipithecines”

- Ventrally orientated foramen magnum
- Flatter face (relatively)
- Still arboreal and found in forested environments
- Curled phalanges
- Possible habitual bipedality but in arboreal setting
- Grasping foot
- Primarily herbivorous/frugivorous
- Loss of honing complex

“Australopithecines”

- Semi-arboreal, semi-cursorial
- Habitual bipeds
- Typical “human” foot
- Curled phalanges
- Primarily herbivorous/frugivorous
- Sagittal ilium
- Curved femurs
- Primitive tool use

Homo

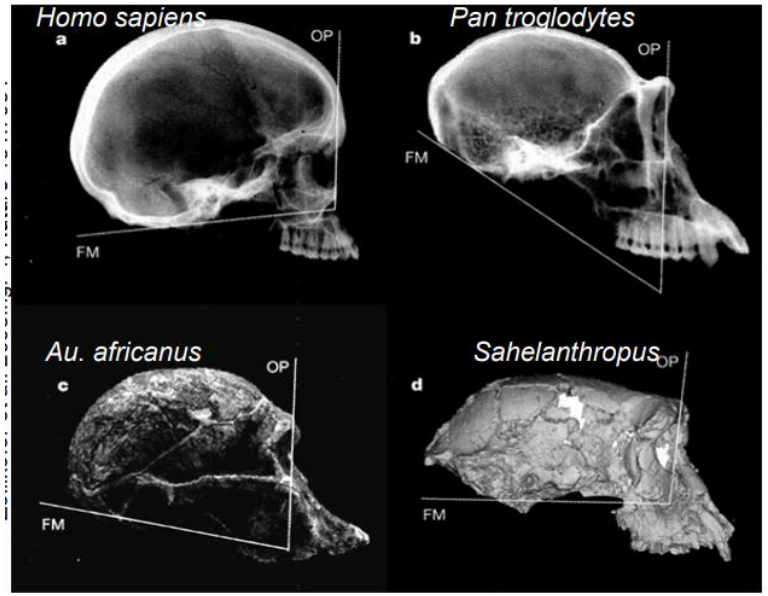
- Habitual bipeds
- Developed shoulder for throwing
- Larger braincase
- Advanced tool use (olduvan, achulean, modern)
- Flatter faces
- Smaller jaws and cheek teeth
- Scavenger/omnivorous diet
- Straight, angled femur
- Straight phalanges
- NOT ARBOREAL – very much cursorial

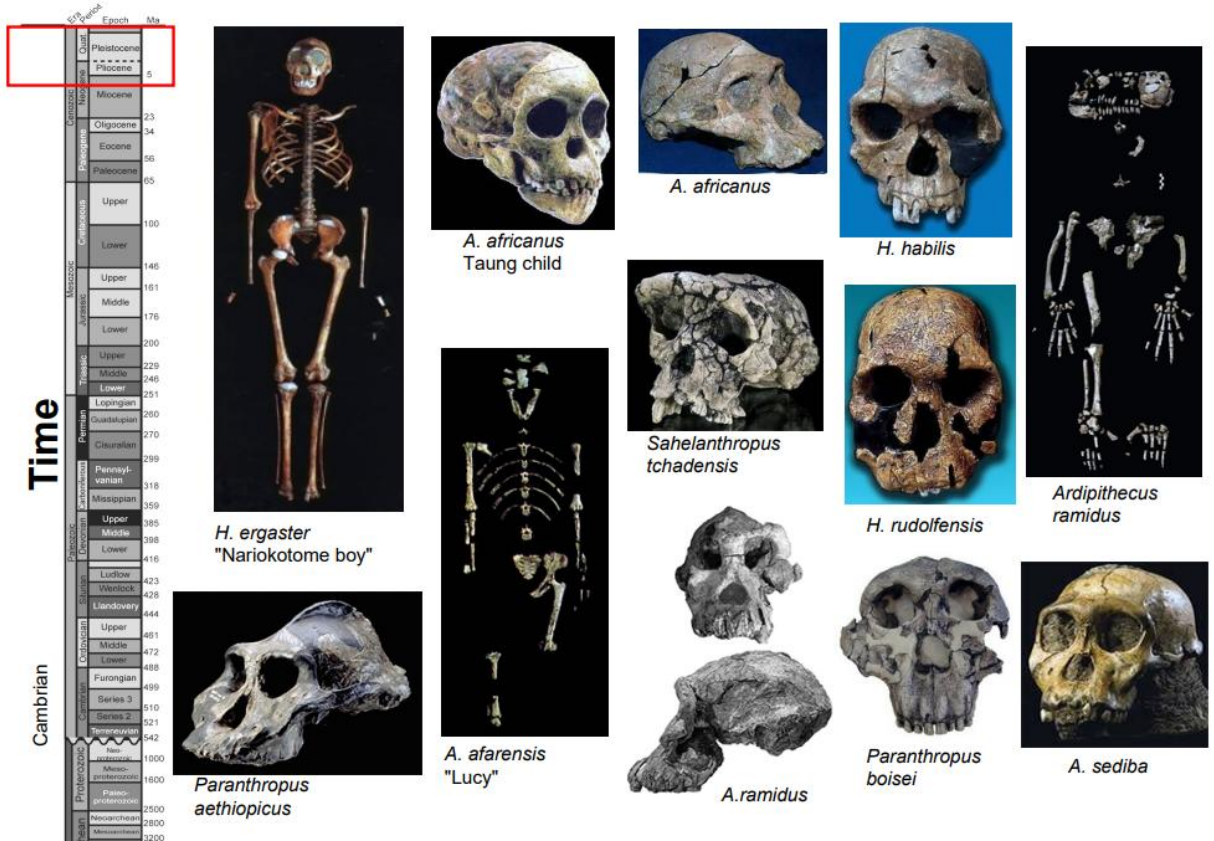
Major defining “Species”

- H. erectus
- H. Sapiens

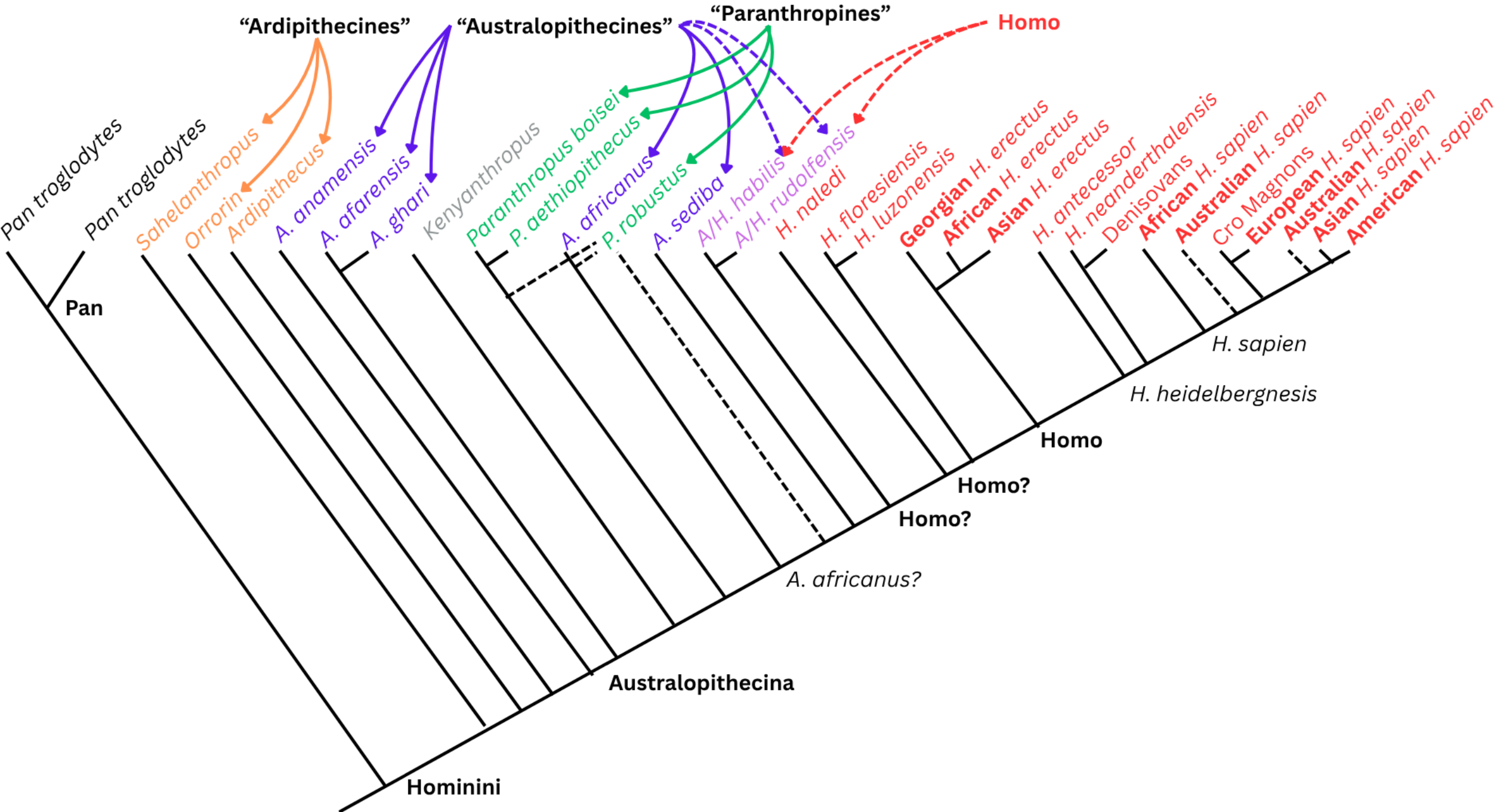
“Paranthropines”

- “Robust” Australopithecines
- Potentially a grade
- Large sagittal crest
- Herbivorous (sedges, nuts, grasses)
- Gave us genital herpes





Hominini


"Ardipithecines"
"Australopithecines"
"Paranthropines"
Homo
Pan troglodytes
Pan troglodytes
Sahelanthropus
Orrorin
Ardipithecus
A. anamensis
A. afarensis
A. ghari
Kenyanthropus
Paranthropus boisei
P. aethiopithecus
A. africanus
P. robustus
A. sediba
A/H. habilis
A/H. rudolfensis
H. naledi
H. floresiensis
H. luzonensis
Georgian H. erectus
African H. erectus
Asian H. erectus
H. antecessor
H. neanderthalensis
Denisovans
African H. sapien
Australian H. sapien
Cro Magnons
European H. sapien
Australian H. sapien
Asian H. sapien
American H. sapien
Pan
H. sapien
H. heidelbergnesis
Homo
Homo?
Homo?
A. africanus?
Australopithecina
Hominini